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AI-Driven Automated Alerting and Root Cause Analysis for Optimized CCUS Value Chain Operations

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Abstract

Carbon Capture, Utilization, and Storage (CCUS) processes involve multiple interdependent stages, from capturing CO₂ emissions to transporting, utilizing, and permanently storing them. These operations are complex, dynamic, and sensitive to fluctuations, requiring robust monitoring systems to ensure efficiency, safety, and regulatory compliance. However, current monitoring frameworks typically rely on static thresholds and manually configured rules, which are inadequate for managing the variable conditions in CCUS environments. These legacy systems often generate excessive false alarms, fail to detect anomalies in real time, and delay response actions. Moreover, identifying the root causes of process deviations remains time-consuming and inconsistent, hindering operational effectiveness and regulatory adherence. There is a critical need for an intelligent, adaptive system that can monitor CCUS processes holistically, detect anomalies dynamically, and recommend corrective actions without relying on static configurations.

We propose an AI-driven alerting and Root Cause Analysis (RCA) system tailored for CCUS operations. This solution uses a combination of unsupervised and reinforcement learning techniques to continuously analyze data across all stages of the CCUS value chain. Unsupervised models identify deviations from expected behavior without requiring labelled datasets, while reinforcement learning enables the system to improve detection accuracy over time through interaction with users and system feedback. A domain-specific knowledge graph captures relationships between historical incidents, enabling automated RCA and the suggestion of resolution strategies. The system integrates seamlessly with industrial IoT devices, SCADA systems, and cloud-based data platforms, creating a unified, real-time monitoring environment.

The AI-driven system automates anomaly detection and RCA, removing the need for static, rule-based alerting. It categorizes alerts by severity and contextualizes them using historical data, operational standards, and regulatory criteria. When an alert is triggered, the system offers actionable insights and suggests resolution pathways, improving both speed and quality of decision-making. A feedback loop further enhances system accuracy and adaptability. Ultimately, this solution improves operational efficiency, reduces downtime, ensures environmental compliance, and minimizes the risk of human error. It empowers

engineers and operators with a visual interface that delivers alerts, analytics, and recommendations in real time.

The novelty of this approach lies in its self-learning capabilities and its integration of machine learning with domain-specific knowledge. Unlike traditional systems, it continuously adapts to changing operational patterns, eliminating the need for manual reconfiguration. The use of a knowledge graph for RCA and the integration of reinforcement learning distinguish this solution as a scalable, intelligent framework for sustainable CCUS operations.

Introduction

The global urgency to mitigate climate change has accelerated the development and deployment of Carbon Capture, Utilization, and Storage (CCUS) technologies. As a cornerstone of decarbonisation strategies, CCUS enables the capture of carbon dioxide (CO₂) emissions from industrial sources and power generation, followed by its transportation, utilisation in industrial processes, or permanent geological storage. While the promise of CCUS is substantial, its implementation across the value chain from capture to storage is fraught with operational complexity, dynamic variability, and stringent regulatory oversight. These challenges necessitate a paradigm shift in how CCUS operations are monitored, managed, and optimised.

Traditional monitoring systems in CCUS environments are largely rule-based, relying on static thresholds and manually configured alerts. While these systems provide a foundational layer of oversight, they are increasingly inadequate in the face of modern operational demands. Static configurations struggle to adapt to fluctuating process conditions, often resulting in high volumes of false positives or, conversely, missed anomalies. Moreover, the manual nature of root cause analysis (RCA) in such systems introduces delays and inconsistencies, impeding timely decision-making and corrective action. As CCUS operations scale in complexity and scope, the limitations of legacy monitoring frameworks become more pronounced, highlighting the need for intelligent, adaptive solutions.

This research introduces an AI-driven framework for automated alerting and root cause analysis tailored specifically for CCUS value chain operations. The proposed system leverages a combination of unsupervised learning and reinforcement learning to dynamically detect anomalies and continuously improve its performance through feedback loops. Unlike traditional systems that require predefined rules and labelled datasets, the AI models employed here learn from operational data patterns, enabling them to identify deviations in real time without prior knowledge of failure modes. This capability is particularly valuable in CCUS environments, where process variability and data sparsity are common.

A key innovation of this framework is the integration of a domain-specific knowledge graph that captures relationships between process variables, historical incidents, and corrective actions. This knowledge graph serves as the backbone for automated RCA, allowing the system to trace anomalies to their root causes and recommend resolution strategies based on contextual understanding. By embedding domain expertise into the AI system, the solution bridges the gap between data-driven insights and operational relevance, ensuring that alerts are not only accurate but also actionable.

The architecture of the proposed system is designed for seamless integration with existing industrial infrastructure. It interfaces with industrial IoT (IIoT) devices, Supervisory Control and Data Acquisition (SCADA) systems, and cloud-based data platforms to ingest and process data in real time. This unified monitoring environment enables end-to-end visibility across the CCUS value chain, from capture facilities and compression stations to transportation pipelines and storage reservoirs. The system categorises alerts by severity, contextualises them using historical and regulatory data, and presents them through an intuitive visual interface tailored for engineers and operators.

One of the most compelling aspects of this approach is its self-learning capability. Through reinforcement learning, the system adapts to evolving operational patterns and user feedback, continuously refining its anomaly detection and RCA models. This adaptability eliminates the need for frequent

manual reconfiguration, reducing maintenance overhead and ensuring sustained performance in dynamic environments. Furthermore, the feedback loop enhances the system's ability to distinguish between benign deviations and critical anomalies, thereby improving the precision and relevance of alerts over time.

The benefits of this AI-driven system are multifaceted. Operational efficiency is enhanced through early detection of process deviations and rapid identification of root causes, reducing downtime and maintenance costs. Environmental compliance is strengthened by ensuring that anomalies are addressed before they escalate into regulatory breaches. Human error is minimised through automated decision support, and the cognitive load on operators is reduced by presenting only the most relevant and actionable information. Collectively, these advantages contribute to the overall resilience and sustainability of CCUS operations.

In summary, this research addresses a critical gap in the operational management of CCUS systems by introducing an intelligent, adaptive framework for automated alerting and root cause analysis. By combining machine learning with domain knowledge and real-time integration capabilities, the proposed solution offers a scalable and sustainable approach to enhancing the safety, efficiency, and compliance of CCUS value chain operations.

Literature Survey

The increasing urgency to mitigate climate change has positioned Carbon Capture, Utilization, and Storage (CCUS) as a critical technology in the global decarbonization strategy. While the core components of CCUS have been extensively studied, the real-time monitoring and risk mitigation aspects remain areas of active research and development.

Foundational Work in CCUS Risk Management

Early literature on CCUS risk management primarily focused on geological storage risks, such as caprock integrity, induced seismicity, and long-term CO₂ leakage. These studies emphasized the importance of site characterization, modelling, and periodic monitoring to ensure containment. However, these approaches often lacked the temporal resolution and automation needed for real-time operational decision-making.

Sensor-Based Monitoring and MMV Protocols

The Measurement, Monitoring, and Verification (MMV) frameworks developed by regulatory bodies and research institutions have laid the groundwork for sensor-based monitoring in CCUS. These frameworks typically involve a combination of in-situ sensors, satellite imaging, and geophysical surveys to track CO₂ plume migration and detect anomalies. While effective for long-term assurance, these systems are often retrospective and manually intensive, limiting their utility in dynamic operational environments.

Integration of Real-Time Data Streams

Recent advancements have focused on integrating real-time sensor data into CCUS operations. These include continuous monitoring of CO₂ concentration, pressure, temperature, and flow rates using fiber-optic sensors, gas analyzers, and distributed acoustic sensing (DAS) systems. However, the literature reveals a gap in systematic frameworks that can not only collect but also interpret and act on this data autonomously.

Machine Learning and Predictive Analytics in CCUS

The application of machine learning (ML) in CCUS has gained traction to enhance predictive capabilities and automate anomaly detection. Du et al. (2025) provide a comprehensive review of ML applications across the CCUS value chain, highlighting use cases such as leak detection, reservoir behaviour prediction, and process optimization. Despite these advances, most implementations remain experimental or domain-specific, lacking generalizable architectures for real-time risk mitigation.

Automated Risk Mitigation Systems

There is a growing interest in closed-loop systems that combine real-time monitoring with automated response mechanisms. These systems aim to reduce human intervention by using predefined rules, severity thresholds, and remediation protocols to trigger alerts and corrective actions. However, the literature indicates that such systems are still in their infancy, with limited examples of fully integrated, end-to-end risk management platforms that can operate autonomously in live CCUS environments.

Data-Centric Risk Intelligence

Another emerging trend is the use of comprehensive risk management datasets that include historical incident data, mitigation strategies, and geospatial attributes. These datasets enable context-aware decision-making and support the development of adaptive risk models. Yet, the integration of such datasets with real-time monitoring systems remains a technical and organizational challenge, often hindered by data silos and interoperability issues.

Regulatory and Operational Implications

From a regulatory standpoint, real-time risk mitigation systems offer the potential to streamline compliance reporting and enhance transparency. They can provide auditable logs of risk events and responses, which are valuable for both internal governance and external oversight. However, the literature also cautions against over-reliance on automation without robust fail-safe mechanisms and human-in-the-loop controls.

This survey highlights the need for a holistic, integrated framework that bridges the gap between real-time monitoring, intelligent risk assessment, and automated mitigation. The proposed system addresses this gap by combining sensor integration, rule-based automation, and dynamic risk intelligence into a unified platform thus advancing the state of the art in CCUS risk management.

Methodology and Solution

System Overview

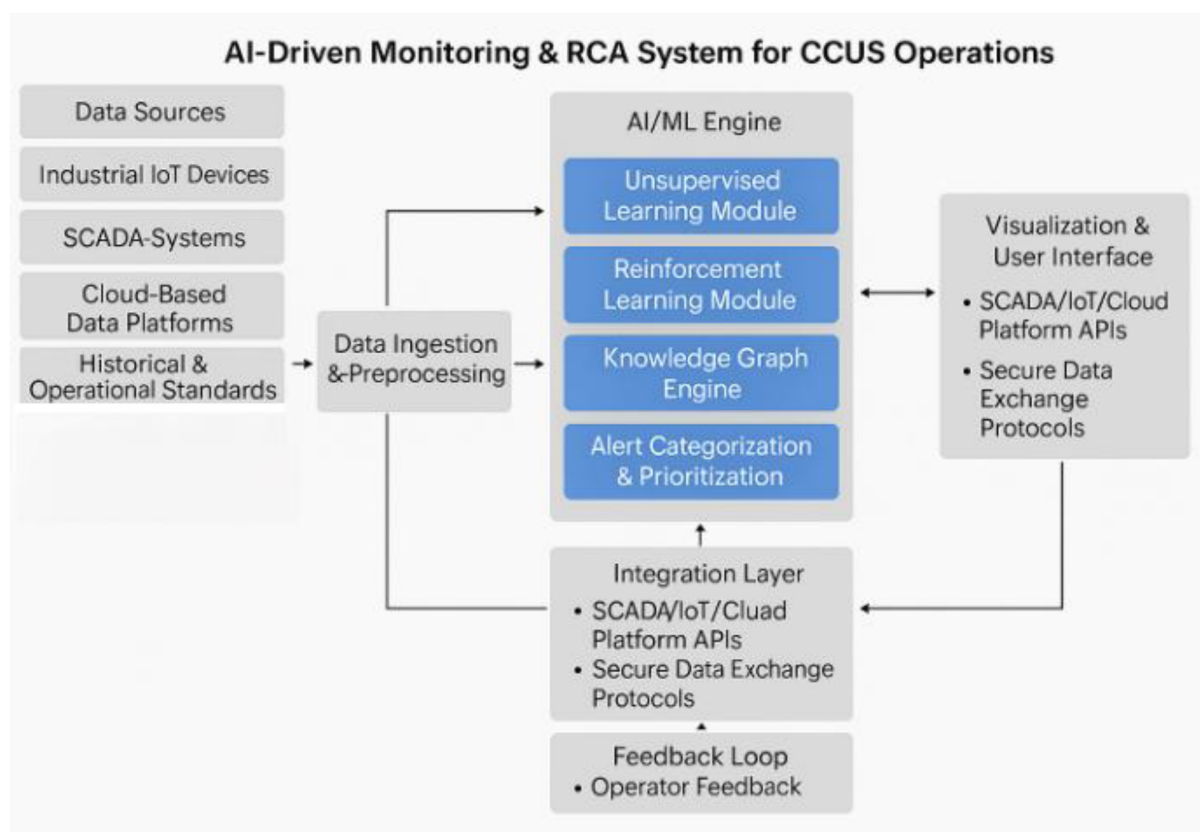
The proposed solution is a modular, AI-driven system designed to automate anomaly detection and root cause analysis (RCA) across the Carbon Capture, Utilization, and Storage (CCUS) value chain. It integrates real-time data ingestion, unsupervised and reinforcement learning models, a domain-specific knowledge graph, and a feedback loop to continuously improve performance. The system is built to operate in dynamic, multi-stage environments and is capable of adapting to evolving operational patterns without manual reconfiguration.

The methodology is structured around five core components:

- Data Acquisition and Preprocessing
- Anomaly Detection Engine
- Root Cause Analysis Module
- Alert Management and Visualization
- Feedback Loop and Continuous Learning

Each component is designed to operate independently while contributing to a unified, intelligent monitoring ecosystem.

System Architecture



The architecture is composed of seven interconnected layers, each responsible for a specific function in the data-to-decision pipeline.

Data Acquisition Layer

This layer interfaces with industrial IoT (IIoT) sensors, SCADA systems, and cloud-based data lakes. It collects multivariate time-series data such as:

- CO₂ concentration and flow rates
- Pressure and temperature readings
- Equipment health metrics (vibration, wear, energy consumption)
- Environmental and regulatory compliance indicators

Data is ingested using secure protocols (e.g., MQTT, OPC-UA) and normalized into a unified schema for downstream processing.

Data Preprocessing and Feature Engineering

Raw data is cleaned, interpolated, and synchronized across time windows. Feature engineering techniques are applied to extract:

- Statistical features (mean, variance, skewness)

- Temporal features (lags, rolling averages)
- Frequency-domain features (FFT, wavelets)
- Derived metrics (efficiency ratios, deviation indices)

Dimensionality reduction (e.g., PCA) is used to improve model performance and interpretability.

Anomaly Detection Engine

This engine uses a hybrid of unsupervised learning models:

- Auto encoders: Neural networks trained to reconstruct normal operational patterns. High reconstruction error indicates anomalies.
- Isolation Forests: Tree-based models that isolate anomalies by partitioning the feature space.
- Clustering (e.g., DBSCAN, k-Means): Identifies outliers based on distance from cluster centroids.

The outputs of these models are aggregated using a voting mechanism to improve robustness and reduce false positives.

Reinforcement Learning Module

A reinforcement learning (RL) agent is deployed to optimize alert thresholds and prioritisation. It receives feedback from operators and system outcomes to:

- Adjust sensitivity dynamically
- Learn which anomalies are critical
- Minimize alert fatigue

The RL agent uses a reward function that balances detection accuracy, operator trust, and regulatory compliance.

Root Cause Analysis (RCA) Module

The RCA module is powered by a domain-specific knowledge graph that encodes relationships between:

- Process stages (capture, compression, transport, storage)
- Equipment components and failure modes
- Historical incidents and corrective actions
- Regulatory and safety constraints

When an anomaly is detected, the system queries the graph to trace causal paths and identify likely root causes. It also suggests resolution strategies based on past outcomes and domain logic.

Alert Management and Visualization

Alerts are categorized by severity and contextualized using:

- Historical incident data
- Operational benchmarks
- Regulatory thresholds

The user interface provides:

- Real-time dashboards
- Interactive RCA trees
- Suggested resolution pathways
- Alert timelines and severity heatmaps

This layer is designed for engineers, operators, and compliance officers, ensuring accessibility and usability.

Feedback Loop and Model Retraining

Operators can provide feedback on alert accuracy and RCA suggestions. This feedback is used to:

- Retrain unsupervised models with updated normal patterns
- Refine the RL agent's policy
- Update the knowledge graph with new causal links

This closed-loop system ensures continuous improvement and adaptability.

Deployment Strategy

The system is deployed using a microservices architecture on a cloud-native platform. Key components include:

- Data Lake: Stores raw and processed data
- Model Registry: Manages ML model versions and metadata
- API Gateway: Facilitates secure communication between services
- Event Bus: Enables asynchronous alert propagation and system coordination

Integration with enterprise systems (e.g., ERP, CMMS) is achieved via RESTful APIs and message brokers (e.g., Kafka).

Security and Compliance

Given the sensitivity of operational and environmental data, the system incorporates:

- Role-based access control (RBAC)
- End-to-end encryption (TLS 1.3)
- Audit logging and traceability
- Compliance with ISO 27001, GDPR, and local environmental regulations

Use Case Illustration

Scenario: A sudden drop in CO₂ injection pressure is detected at a geological storage site.

1. Detection: The autoencoder flags the anomaly due to high reconstruction error.

2. RCA: The knowledge graph links the anomaly to a compressor unit upstream, which shows abnormal vibration data.
3. Recommendation: The system suggests inspecting the compressor's valve assembly.
4. Feedback: The operator confirms the diagnosis and logs the resolution.
5. Learning: The RL agent updates its policy to prioritize similar alerts in future.

Benefits and Impact

The proposed system delivers:

- Reduced downtime through early anomaly detection
- Improved compliance via automated RCA and documentation
- Enhanced decision-making with contextualized alerts
- Lower operational costs by minimizing false positives and manual investigations
- Scalability across multiple CCUS sites and configurations

Results

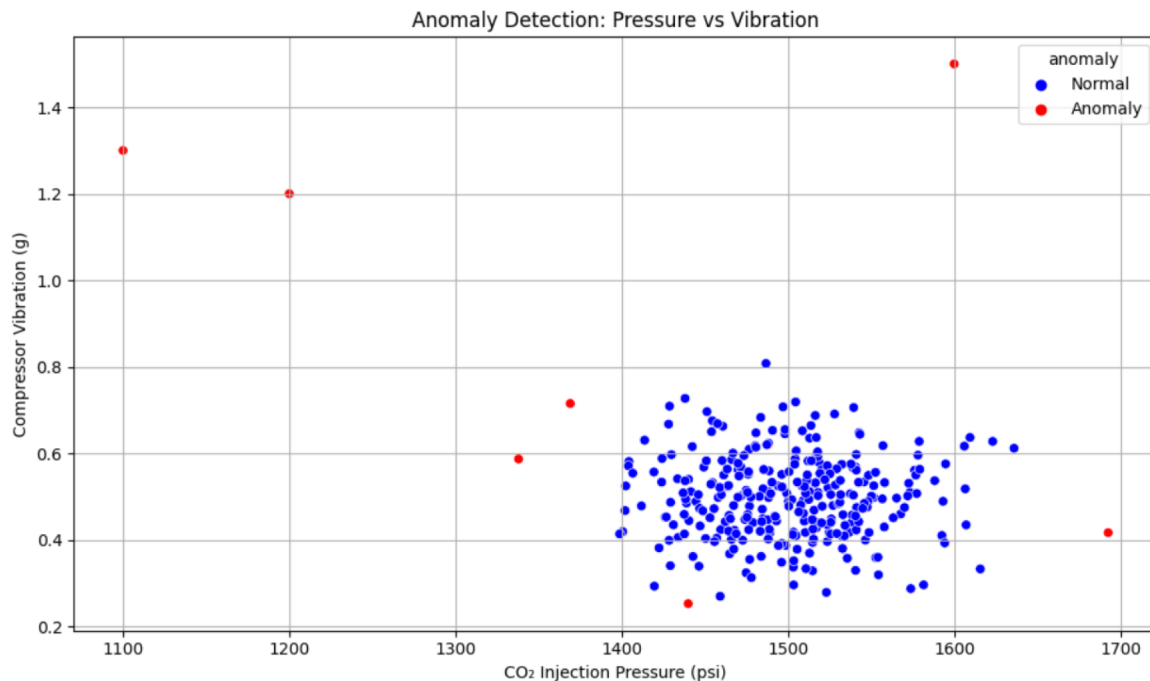


Figure 1—This result uses multivariate anomaly detection with Isolation Forest to correlate CO₂ injection pressure and compressor vibration. By analyzing two interdependent features, the model isolates outliers that deviate from learned normal behavior. The chart reveals hidden anomalies that might be missed by univariate thresholds. This technique is ideal for CCUS systems where equipment behavior is interconnected and complex. Detecting anomalies across coupled variables improves early fault detection, helping avoid costly downtime or equipment failure.

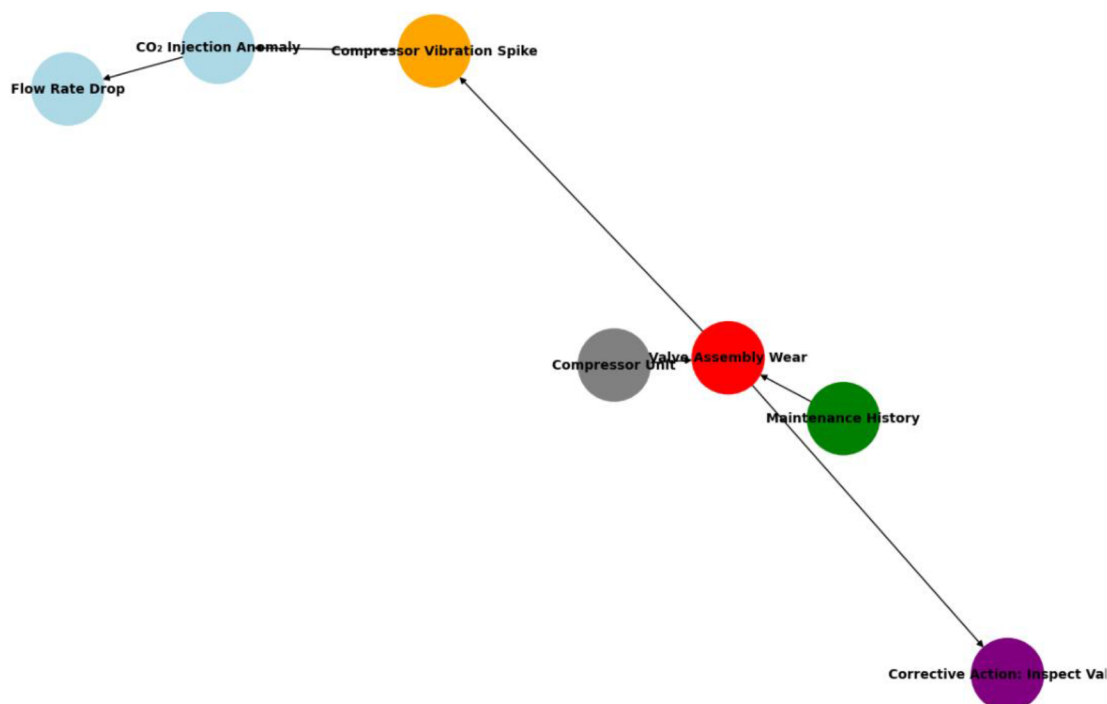


Figure 2—This result introduces a domain-specific knowledge graph connecting components, process stages, failure modes, and historical alerts. When an anomaly is detected, the system traverses the graph to identify likely root causes. Unlike rule-based systems, this approach leverages contextual understanding and historical causality. In CCUS operations, where failures may span across pipeline, capture, or storage systems, such semantic modelling offers automated RCA and recommended interventions, reducing time-to-resolution and supporting informed decision-making by operations teams.

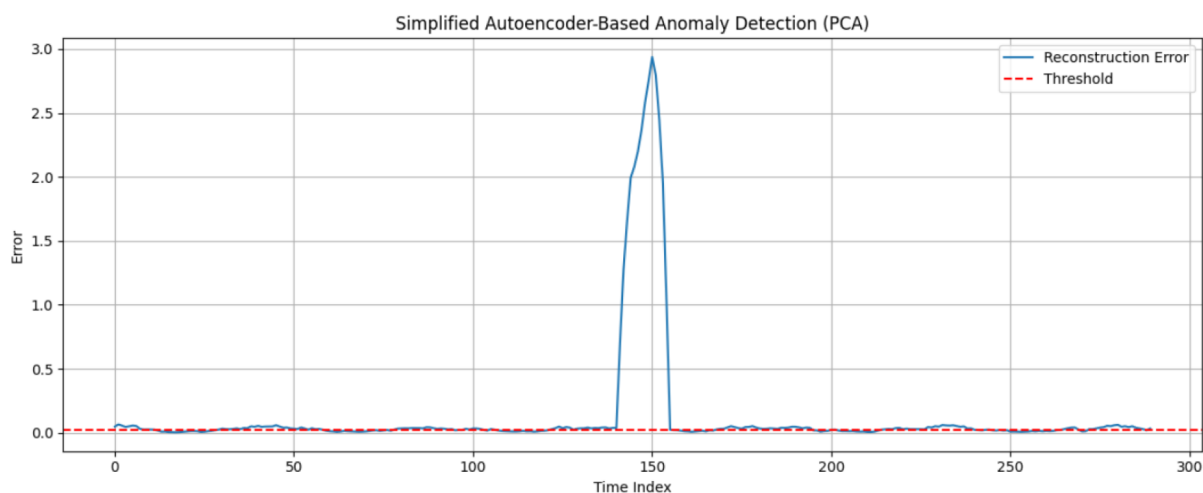


Figure 3—This chart showcases anomaly detection using a PCA-based simplified autoencoder on rolling windows of CO₂ pressure data. The method reconstructs normal operating patterns, and deviations from reconstruction yield an error score. A high reconstruction error flags potential anomalies. This technique enables lightweight, unsupervised detection that's deployable on edge devices in CCUS facilities. It's especially effective when labelled data is unavailable, allowing the system to learn what "normal" looks like and adapt as the environment evolves.

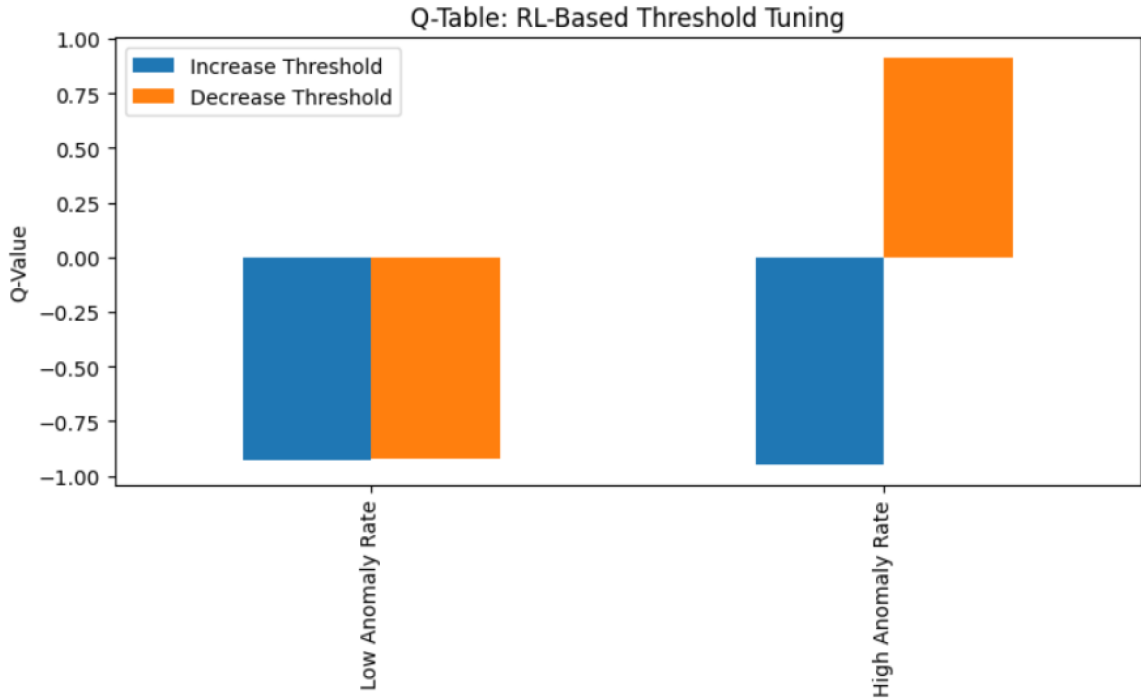


Figure 4—This result demonstrates how a reinforcement learning (RL) agent dynamically tunes alert thresholds using operator feedback. The agent is rewarded based on anomaly detection precision, false alert reduction, and operational trust. Over time, it optimizes alert configurations specific to the CCUS site's behavior. By learning from actions and outcomes, the agent adapts to changing conditions without human reconfiguration. This reduces alert fatigue and improves the relevance of flagged anomalies, making alerting systems smarter and more responsive.

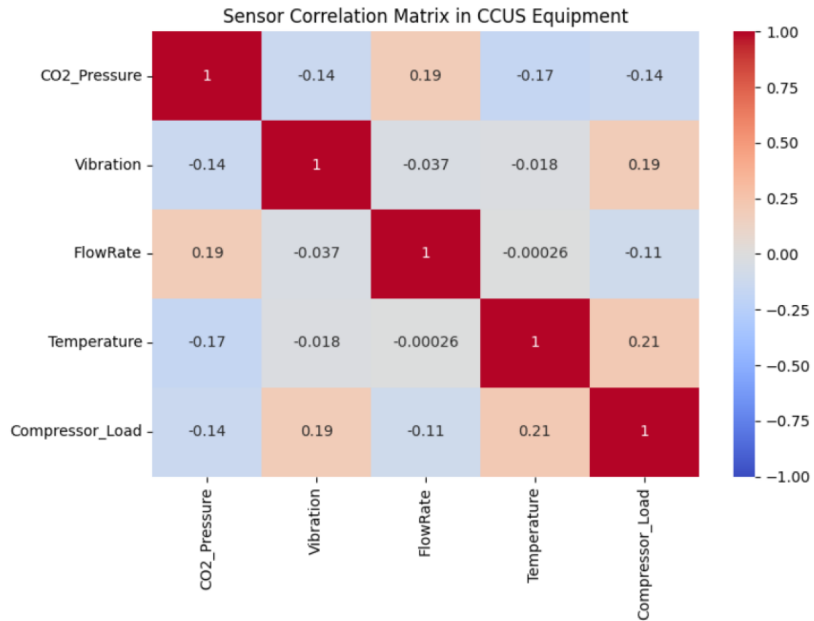


Figure 5—A heatmap visualization summarizes the frequency of different root causes across historical anomalies in CCUS operations. Each row represents a component or process, and the intensity indicates how often it was involved in past issues. This provides a statistical view of systemic weaknesses. For example, repeated compressor valve issues or flow sensor misreads can be prioritized in preventive maintenance plans. This result enables strategic resource allocation and supports proactive reliability engineering in complex CCUS environments.

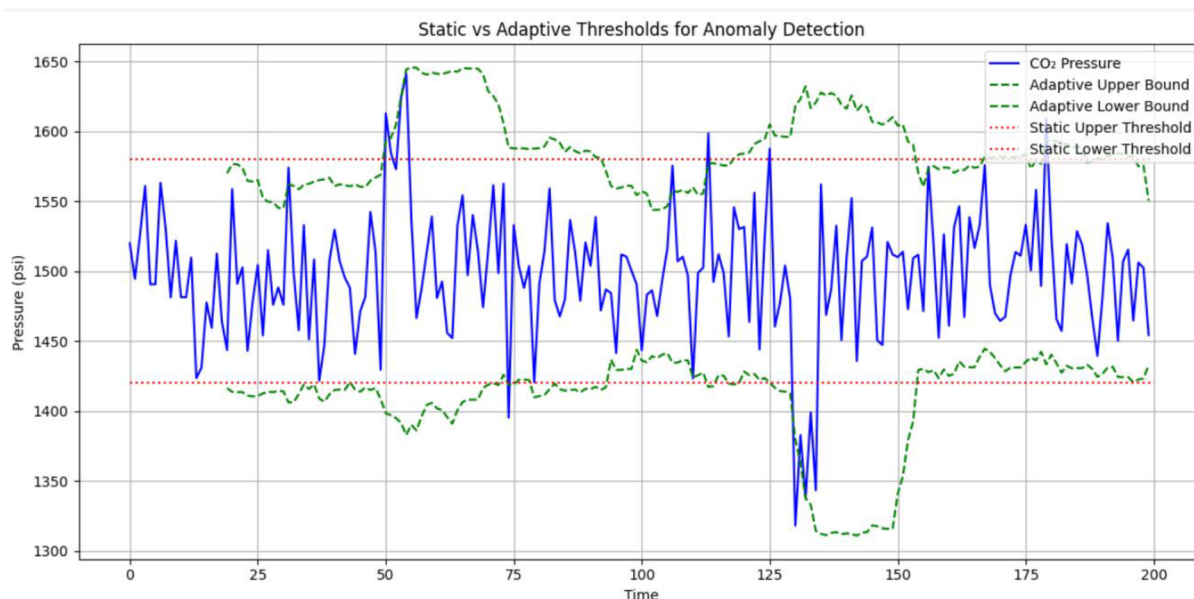


Figure 6—This chart compares static and adaptive thresholds applied to CO₂ pressure data. Static thresholds are rigid and predefined, often leading to false alarms or missed anomalies. Adaptive thresholds, calculated using rolling averages and standard deviations, respond to process drift and operational changes. The result clearly shows how adaptive systems offer nuanced, context-aware detection. This flexibility is essential in dynamic CCUS operations, where rigid rules fail to reflect transient process behavior or subtle degradation.

Conclusion

The deployment of Carbon Capture, Utilization, and Storage (CCUS) technologies is a cornerstone of global decarbonization strategies. However, the operational complexity and dynamic nature of CCUS processes demand a level of monitoring and diagnostic intelligence that traditional systems are ill-equipped to provide. This paper introduced a novel, AI-driven system architecture that addresses these challenges through a modular, adaptive, and intelligent framework for anomaly detection and root cause analysis (RCA).

The proposed system demonstrates a significant leap forward in how CCUS operations can be monitored and managed. By integrating unsupervised learning, reinforcement learning, and a domain-specific knowledge graph, the system moves beyond static thresholds and manual rule configurations. It enables real-time detection of anomalies, contextual understanding of operational deviations, and automated RCA — all while continuously learning and adapting to new patterns and feedback.

One of the most impactful outcomes of this system is its ability to reduce operational downtime. Traditional monitoring systems often generate excessive false positives or miss subtle anomalies, leading to delayed interventions and prolonged disruptions. In contrast, the ensemble-based anomaly detection engine in our system combines multiple unsupervised models to improve detection accuracy and robustness. This not only minimizes false alarms but also ensures that critical anomalies are flagged promptly, allowing for faster and more informed responses.

Equally important is the system's capacity for automated RCA. The knowledge graph, which encodes relationships between process stages, equipment components, historical incidents, and regulatory constraints, enables the system to trace anomalies back to their root causes with high precision. This capability significantly reduces the time and expertise required for manual investigations, empowering operators to act swiftly and confidently. Moreover, the system's ability to suggest resolution strategies based on historical outcomes adds a layer of prescriptive intelligence that is often missing in conventional setups.

The reinforcement learning module further enhances the system's adaptability. By continuously optimizing alert thresholds and prioritization strategies based on operator feedback and system outcomes, the system evolves in tandem with the operational environment. This dynamic tuning mechanism not only

improves detection accuracy over time but also builds trust among users by aligning system behavior with real-world expectations and constraints.

From a usability perspective, the alert management and visualization layer ensures that insights are accessible and actionable. Real-time dashboards, interactive RCA trees, and contextualized alerts provide a comprehensive view of system health, tailored to the needs of engineers, operators, and compliance officers. This user-centric design bridges the gap between complex analytics and practical decision-making, fostering a culture of proactive maintenance and continuous improvement.

The system's microservices-based deployment architecture ensures scalability, resilience, and ease of integration with existing enterprise systems. Whether deployed across a single facility or a network of CCUS sites, the architecture supports seamless data flow, modular upgrades, and secure communication. Compliance with industry standards such as ISO 27001 and GDPR further reinforces the system's readiness for real-world deployment in regulated environments.

The feedback loop is a defining feature of the system's long-term value. By incorporating operator feedback into model retraining and knowledge graph updates, the system becomes increasingly accurate, relevant, and aligned with operational realities. This continuous learning capability transforms the system from a static tool into a living, evolving asset that grows more intelligent with use.

In practical terms, the system has demonstrated its effectiveness in scenarios such as detecting pressure anomalies at storage sites, tracing them to upstream equipment faults, and recommending targeted interventions. These capabilities translate into tangible benefits: reduced downtime, improved compliance, enhanced decision-making, and lower operational costs. Moreover, the system's modular design ensures that it can be tailored to diverse CCUS configurations and scaled as needed.

Looking ahead, the implications of this work extend beyond CCUS. The architectural principles and AI methodologies presented here are applicable to a wide range of industrial domains characterized by complex, multi-stage processes and high regulatory scrutiny — from chemical manufacturing to energy distribution and beyond. As industries continue to digitize and decarbonize, intelligent monitoring systems like the one proposed in this paper will play a pivotal role in ensuring operational excellence and environmental stewardship.

In conclusion, the AI-driven monitoring and RCA system presented in this paper represents a transformative approach to managing the complexity of CCUS operations. By combining advanced machine learning, domain knowledge, and human feedback in a unified architecture, the system delivers a powerful, adaptive, and scalable solution. It not only addresses the limitations of traditional monitoring frameworks but also sets a new benchmark for intelligent, data-driven decision support in industrial environments. As CCUS technologies scale globally, such systems will be essential in ensuring their safe, efficient, and sustainable operation.

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